Find the first movement in a tower

Task 2.3 / Prepare and draw

Name _____ Team _____ Date _____

01 Set up the investigation

Agree on height, allowed stock, and a stop rule with your teacher. Draw the tower's base and bracing. Mark the joint you predict will move first.

Measure the base and tower height. Record the starting lean or displacement.

Build target	Limit and unit	As built	Partner check
Height			
Footprint			
Stock / tape			

02 Plan it on paper

Sketch the bracing. Predict which joint or member will move first and how you will see it.

My prediction, chosen feature or design limit

Find the first movement in a tower

Task 2.3 / Test, compare and explain

Name _____ Team _____ Date _____

03 Run the test and record what happened

Build a freestanding tower to the recorded height using the class material limit. Have a partner check dimensions before loading. Load in equal increments under teacher supervision. Stop at the first major lean or joint slip. Photograph or sketch that first failure.

Increment	Total load and unit	Lean / displacement	First change or failure
0			
1			
2			
3			
4			

04 Use the reading

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Complete the assigned checks in the interactive reader. Here, identify a page and one idea that helps explain your result or design decision.

05 Individual explanation

Identify the first failure and the load or trial when it appeared. What one change follows from it?

Keep these sheets with the work. Save the individual explanation in the lesson’s Explain step.